A

TECHNICAL REPORT

ON

ANALYSIS OF USER REVIEWS AND EXPERIENCE PATTERNS FOR PRODUCT IMPROVEMENT

WRITTEN BY

DATA ANALYTICS TEAM

(CAPSTONE TEAM 5)

FOR THE

CAPSTONE PROJECT

INTRODUCTION

The rapid growth of digital platforms has intensified competition among applications, making user experience, trust, and satisfaction critical success factors. User reviews provide direct insight into how users perceive app performance, usability, and reliability. When properly analyzed, these reviews reveal patterns that ratings alone cannot explain.

This project focuses on analyzing reviews from existing applications with similar functionality to a proposed Skill Swap web application. By learning from real user experiences across comparable platforms, the analysis helps identify common frustrations, value drivers, and behavioral signals that influence satisfaction, recommendation, and churn. These insights can be leveraged to design a better, more user-centered application.

PROJECT OVERVIEW

This project analyzes user reviews from platforms similar in purpose to a Skill Swap application, with the goal of transforming unstructured feedback into actionable business intelligence.

Microsoft Excel was used for data collection, trimming, and transformation, while Microsoft Power BI was used for analysis, DAX calculations, and interactive dashboard visualization. The project focuses on understanding how user sentiment, feature experiences, and reported issues influence satisfaction, recommendation behavior, and churn risk.

The outcome of this analysis is a set of insights that can guide product design decisions, feature prioritization, and customer retention strategies for a new application.

PROBLEM STATEMENT

Many digital platforms experience low user satisfaction and high churn due to unresolved usability issues, trust concerns, payment problems, and performance challenges. These issues are often buried within unstructured review text and scattered ratings, making them difficult to analyze without proper structuring.

Without a systematic approach to analyzing user feedback, product teams struggle to understand what truly drives dissatisfaction, which features act as deal-breakers, and what factors encourage users to recommend or abandon an app. This project addresses that gap by structuring and analyzing review data to support data-driven product improvement.

- **Project Goal**

  The goal of this project is to understand and predict user satisfaction, recommendation behavior, and churn risk by analyzing app review data. The analysis aims to identify key factors driving negative user experiences and uncover actionable insights that can be used to improve product performance, user trust, and customer retention for a proposed Skill Swap application.

DATASET OVERVIEW

The dataset used for this project was compiled from user reviews collected across multiple online platforms where users actively share feedback about digital products. Review sources include Google Play Store, Reddit, Quora, Trustpilot, and Google Reviews.

The reviews focus on platforms with functionality similar to a Skill Swap application, including Upwork, Fiverr, Freelancer, Glassdoor, and LinkedIn. These platforms were selected because they share comparable user journeys, trust dynamics, and service models.

Data collection was carried out collaboratively by the capstone team. Platforms and applications were shared among team members to avoid duplication. Individual datasets were then submitted and centrally collated into a single dataset.

In total, 14 review files were collected. After consolidation, trimming, cleaning, and deduplication, the final dataset contained 191 distinct user reviews, structured to support sentiment analysis, feature-level feedback analysis, issue identification, and churn-related insights.

Data Structure

The final dataset is structured to enable comprehensive analysis of user sentiment, feature experiences, and churn risk. Key columns include:

- Reviewer_ID: Unique identifier for each reviewer.
- Review_Date: Date the review was submitted, standardized to YYYY-MM-DD.
- Sentiment: Overall sentiment of the review (Positive, Negative, Neutral).
- Mentioned_Feature_1: First feature referenced in the review.
- Mentioned_Feature_2: Second feature referenced in the review.
- Mentioned_Feature_3: Third feature referenced in the review.
- Mentioned_Feature_4: Fourth feature referenced in the review.
- Issue_Flag_Payment: Indicator if the review reported payment issues (Positive/Negative).
- Issue_Flag_Trust: Indicator if the review reported trust concerns (Positive/Negative).
- Issue_Flag_UX: Indicator if the review reported UX/UI issues (Positive/Negative).
- Issue_Flag_Performance: Indicator if the review reported performance issues (Positive/Negative).

- Satisfaction_Score: Derived metric representing overall user satisfaction.
- Recommendation_Likelihood: Derived metric capturing the likelihood a user will recommend the platform.
- Churn_Risk: Derived metric estimating the risk of a user discontinuing usage based on sentiment, issue flags, and satisfaction.

TOOLS AND TECHNOLOGIES

MICROSOFT EXCEL – DATA COLLECTION AND TRANSFORMATION

- **Data Collation and Consolidation**
  Microsoft Excel was used as the primary tool for collating and transforming the dataset. All collected review files were stored in a shared drive and consolidated into a master worksheet named Data. This created a single structured environment for assessment and cleaning.

- **Dataset Trimming and Relevance Filtering**
  The dataset was reviewed to identify non-essential and low-value columns. Irrelevant fields were removed to trim the dataset, ensuring that only variables relevant to sentiment analysis, feature evaluation, issue identification, and churn analysis were retained. This reduced noise and improved analytical clarity.

- **Data Quality Checks and Deduplication**
  The dataset was examined for missing values and duplicate records, particularly in key independent variables such as Reviewer Name and Reviewer ID. Duplicate records were removed using Excel's Remove Duplicates functionality. Each remaining row was treated as a distinct review, resulting in a final dataset of 191 unique reviews. A backup copy of the cleaned data was saved as Final Data.

- **Structuring and Standardization**
  The cleaned dataset was converted into an Excel Table to enable structured referencing and filtering. The Review Date column was standardized into a consistent date format to support time-based analysis. Columns with little analytical value, such as Review Link, were removed.

- **Feature Categorization for Analysis**
  The Mentioned Feature field contained multiple feature values in some rows. These were split into multiple Feature Category columns using Excel's Text to Columns functionality. This categorization enabled clearer aggregation and visualization of feature-level feedback in Power BI.

- **Issue Flag Normalization**
  Issue-related fields contained mixed formats, with some recorded as Positive/Negative and others as binary values (0 and 1). All issue flags were standardized into a single Positive/Negative format to ensure consistency, accurate aggregation, and clean KPI calculations.

- **Final Formatting and Documentation**
  Basic formatting was applied for readability and consistency. Screenshots of both the raw and cleaned datasets were captured to document the transformation process and ensure transparency.

MICROSOFT POWER BI – ANALYSIS AND DASHBOARD VISUALIZATION

- **Data Import:**
  The cleaned Excel dataset (Final Data) was imported into Power BI, and the relevant sheet was selected for analysis.
- **Data Preview in Power Query:**
  The dataset was reviewed in Power Query to check column quality, distribution, and profiles. Empty rows were identified, and distinct and unique values were reviewed to understand data completeness.
- **Feature Transformation:**
  The Mentioned_Feature_1 to Mentioned_Feature_4 columns were unpivoted to better analyze feature-level feedback. A duplicate table was created, unnecessary columns were removed, and the feature columns were unpivoted into a single Mentioned_Feature column. Empty rows in issue flag columns were preserved to allow accurate aggregation.
- **Data Modeling:**
  After transformations, a many-to-one relationship was created between the main table and the unpivoted feature table using the Reviewer_ID column. This allowed feature-level insights to roll up correctly into the main dataset.
- **DAX Measures:**
  Key measures were created using DAX to support analysis, including sentiment distribution, satisfaction score, recommendation likelihood, churn risk, and issue prevalence.
- **Visualization and Report Design:**
  The Power BI report was structured to tell a clear story:
  - **Cover Page:** Introduces the project, objectives, and dataset.
  - **Report Pages (Overview and Insight): The** Overview Page summarizes high-level metrics, and the Insight Page provides detailed feature-level analysis and issue trends.
  - **Pre-Analysis Page:** Shows metrics and distributions before detailed exploration, providing context.
  - **Observation Page:** Highlights patterns and key findings from the analysis.
  - **Recommendation Pages (2 pages):** Presents actionable suggestions based on insights from the report and observation pages.

Interactive filters, slicers, and conditional formatting were applied throughout the report to ensure clarity and interactivity.

Key DAX Measures

Several DAX measures were created to quantify user sentiment, satisfaction, and behavior:

- Average Churn Risk: Calculates the average churn risk across all reviews.
- Least Recommended App: Identifies the app with the lowest recommendation rate.
- Goal Achievement Rate (%): Measures the proportion of users who reported achieving their goals.
- Most Criticized App: Highlights the app with the highest number of negative reviews.
- Sentiment Distribution: Shows the proportion of positive, neutral, and negative reviews.
- Churn Rate by Issue: Measures the churn rate associated with specific issues.
- Churn Risk Rate: Calculates overall risk of churn across segments or features.

Sample DAX Formulas:

```
Average Churn Risk =
AVERAGE(ReviewsData[Churn Risk])


Least Recommended App =
VAR RecTable =
   ADDCOLUMNS(
      SUMMARIZE(ReviewsData, ReviewsData[App Name]),
      "RecommendRate",
      DIVIDE(
         CALCULATE(COUNTROWS(ReviewsData), ReviewsData[Would Recommend] = "Yes"),
         COUNTROWS(ReviewsData)
      )
   )
RETURN
   MAXX(
      TOPN(1, RecTable, [RecommendRate], ASC),
      ReviewsData[App Name]
   )


Goal Achievement Rate (%) =
DIVIDE(
   CALCULATE(
      COUNTROWS(ReviewsData),
      ReviewsData[Goal Achieved] = "Yes"
   ),
   COUNTROWS(ReviewsData)
)
```

METHODOLOGY

This project followed a structured workflow to turn unstructured user reviews into actionable insights:

- **Data Collection and Preparation: Reviews** were gathered from multiple online platforms and consolidated. Data cleaning ensured completeness, removed duplicates, standardized dates, and structured feature mentions and issue flags.
- **Data Structuring and Modeling:** The dataset was organized to support feature-level and overall analysis. Relationships were defined between variables, and derived metrics like satisfaction score, recommendation likelihood, and churn risk were calculated.
- **Exploratory Analysis and Pre-Assessment:** Overall trends in sentiment, satisfaction, and issue prevalence were examined before detailed analysis. This pre-analysis stage helped identify patterns and guide deeper exploration.
- **Insight Generation:** Analysis focused on how feature experiences, issues, and sentiment affected satisfaction and churn. Observations were made at both feature and aggregated levels to uncover actionable insights.
- **Recommendations Development:** Insights were translated into actionable recommendations for product improvements, feature prioritization, and customer retention.
- **Reporting and Presentation:** The findings were presented in a multi-page report with a cover page, report pages (overview and insight), pre-analysis and observation pages, and recommendation pages, guiding stakeholders from high-level trends to actionable decisions.

DATA STORY

The data narrates how users interact with and experience digital platforms, highlighting:

- What users' value?
- What frustrates them?
- How these experiences shape satisfaction, trust and loyalty decisions

This story-driven approach ensures insights are actionable and business-relevant.

a. Key Analytical Questions

1. Sentiment Analysis

- What is the relationship between rating and sentiment?
- How does sentiment vary by feature category (UX, Payment and Trust)?
- Are there cases of low ratings with positive sentiment or high ratings with negative sentiment?

2. Feature Impact Analysis

- Which features generate the most negative reviews?
- Which feature categories contribute most to churn risk?

3. Issue Analysis

- Frequency of Payment, Trust, UX, and Performance issues
- Impact of each issue type on rating, sentiment and satisfaction
- Which issues drive the lowest ratings?

4. Customer Retention Analysis

- Common complaints among high churn-risk users.
- Features most strongly linked to churn.
- How trust and UX issues affect recommendation behavior.

5. Feature-Level Feedback Analysis

- Most frequently mentioned features.
- Feature-level sentiment distribution.
- Feature impact on ratings and satisfaction.

6. Trust & Payment Risk Analysis

- Impact of trust issues on recommendation and churn.
- Effect of payment issues on sentiment and satisfaction.

DATA COLLECTION AND TRANSFORMATION

1. Collection of all secondary data from the Capstone Team 5 Data Analytics Team into a drive named Data Collection and Transformation Drive. There were 14 file reviews collected from Playstore, Google, Mobile App, Mobile app & web, Reddit, Trustpilot for Upwork, Fiverr, Freelancer, Glassdoor, Quora, and LinkedIn, which are existing projects similar to our Skill Swap Web App.
2. A brain storm session was conducted to determine the tool to be used to join the data. SQL, Python and Excel were suggested. Excel was opted for easier interactions with other members.
3. A worksheet was opened and all reviews from the 14 different files were copied and pasted on the worksheet named Data.
4. The Data was observed for duplicates and blank cells especially in independent variables such as Reviewer ID and Reviewer Name.
5. Duplicates were found in the Reviewer name, thereby removed using the "Remove Duplicates" function on the Data tab and Data tools Ribbon.

6. Duplicates were found in Reviewer ID. However, every row was made distinct and this brought total reviews to a Count distinct total of 191 reviews.
7. The data was duplicated to “Final Data” for back up reasons.
8. Data was structured to a table using Ctrl T.
9. Using the Filter button, consistency was checked across independent variables.
10. Date column was formatted to date format.
11. Review Link column was blank therefore removed.
12. Standardization was done, where some categories were categorized in the mentioned feature column.
13. For a dependent variable, Feature Category Column that has multiple answers, text to columns was used to split the answers to 4 feature category columns.
14. The font was formatted to Calibri. Font size 9.
15. Some of the mentioned features were similar, therefore categorised for easy identification and visualization.
16. The TRIM function was used to remove leading and trailing spaces, then misspelt words within the dataset were corrected.

Predictive Modeling Opportunities

Potential models that can be developed from this dataset include:

- Churn Risk Prediction using issue flags and sentiment.
- User Satisfaction Prediction using sentiment and feature categories.
- Recommendation Prediction based on ratings, trust, and UX indicators.

These models support proactive decision-making and retention strategy design.

Review ID	Reviewe	App Na	Platforr	Review Date	Review	Rating	Full Rev	Sentime	Mentio	Feature	Paymen	Trust Is	UX Issue	Perform	Goal Ac	User Sat	Would I	Churn R
RVW_001	Grace Thc	Fiverr	Play Store	23/06/2024		3	User feed	Negative	Customer	UX			Negative		0	Low	No	High
RVW_002	Linda Per	Fiverr	Play Store	26/06/2024		5	User feed	Negative	Customer	UX			Negative		0	Low	No	High
RVW_003	Aisha Law	Fiverr	Play Store	02/12/2024		1	User feed	Negative	Account S	Trust		Negative			0	Low	No	High
RVW_004	Aisha Law	Fiverr	Play Store	18/08/2024		3	User feed	Neutral	Gig Uploa	Performance					0	Medium	Maybe	Medium
RVW_005	Samuel Be	Fiverr	Play Store	25/07/2024		5	User feed	Neutral	Gig Uploa	Performance					0	Medium	Maybe	Medium
RVW_006	Peter Johr	Fiverr	Play Store	30/04/2024		4	User feed	Positive	Onboardi	UX					1	High	Yes	Low
RVW_007	Fatima M	Fiverr	Play Store	05/06/2024		2	User feed	Negative	Account S	Trust		Negative			0	Low	No	High
RVW_008	John Ade	Fiverr	Play Store	06/05/2025		5	User feed	Positive	Push Noti	Notifications / UX					1	High	Yes	Low
RVW_009	Daniel Ok	Fiverr	Play Store	26/04/2024		5	User feed	Positive	App Stabi	Performance					1	High	Yes	Low
RVW_120	Kelvin Ob	Fiverr	Play Store	24/11/2024		3	User feed	Neutral	Order Car	Payment / Trust					0	Medium	Maybe	Medium
RVW_001	u/techsav	Fiverr	Reddit	2024-01-12		4	Used Five	Positive	Gig delive	Service	0	0	0	0	Yes	High	Yes	Low
RVW_002	u/startup	Fiverr	Reddit	2024-02-03		3	Fiverr is g	Neutral	Seller qua	Marketpla	0	0	0	0	Partial	Medium	Maybe	Medium
RVW_003	u/anon_d	Fiverr	Reddit	2024-02-18		2	Had issue	Negative	Customer	Support	1	1	0	0	No	Low	No	High
RVW_004	u/devtalk	Fiverr	Reddit	2024-03-01		5	Great exp	Positive	Milestone	Project M	0	0	0	0	Yes	High	Yes	Low
RVW_005	u/uxcritic	Fiverr	Reddit	2024-03-10		3	The app w	Neutral	Interface	UX	0	0	1	0	Yes	Medium	Maybe	Medium
RVW_006	u/freelan	Fiverr	Reddit	2024-03-22		4	Payment p	Positive	Payment p	Trust & Sa	0	0	0	0	Yes	High	Yes	Low
RVW_007	u/compla	Fiverr	Reddit	2024-04-05		1	Account g	Negative	Account n	Trust & Sa	0	1	0	0	No	Low	No	High
RVW_008	u/content	Fiverr	Reddit	2024-04-14		5	Perfect fo	Positive	Gig marke	Marketpla	0	0	0	0	Yes	High	Yes	Low
RVW_009	u/bugrep	Fiverr	Reddit	2024-04-27		2	Experienc	Negative	App stabi	Performa	0	0	0	1	No	Low	No	High
RVW_010	u/reviewr	Fiverr	Reddit	2024-05-02		4	Reviews a	Positive	Ratings sy	Trust & Sa	0	0	0	0	Yes	High	Yes	Low
RVW_011	u/sidehus	Fiverr	Reddit	2024-05-11		3	Fees can a	Neutral	Service fe	Pricing	0	0	0	0	Partial	Medium	Maybe	Medium
RVW_012	u/brandb	Fiverr	Reddit	2024-05-19		5	Found a l	Positive	Talent dis	Marketpla	0	0	0	0	Yes	High	Yes	Low
RVW_013	u/securep	Fiverr	Reddit	2024-06-01		2	Refund pr	Negative	Refunds	Payments	1	0	0	0	No	Low	No	High
RVW_014	u/remotel	Fiverr	Reddit	2024-06-10		4	Easy to fir	Positive	Search fil	UX	0	0	0	0	Yes	High	Yes	Low
RVW_015	u/researc	Fiverr	Reddit	2024-06-21		3	Decent pl	Neutral	Seller vett	Trust & Sa	0	0	0	0	Partial	Medium	Maybe	Medium
RVW_016	David Mil	Freelance	Quora	2023-05-28		1	My experi	Negative	Skill exch	Collabora	Positive	Positive			0	Low	No	High
RVW_017	Daniel Kir	Freelance	Quora	2023-04-03		3	I tried Fre	Neutral	Skill swap	Collabora	Positive	Positive	Positive	Negative	1	Medium	No	Medium
RVW_018	Maria Go	Freelance	Quora	2025-05-17		5	I have bee	Positive	Skill exch	Collabora	Negative	Positive			1	High	Yes	Low
RVW_019	Yusuf Abd	Freelance	Quora	2024-12-10		1	My experi	Negative	Skill swap	Project-Ba	Negative	Negative	Positive		0	Low	No	High
RVW_020	Samuel O	Freelance	Quora	2024-05-01		1	My experi	Negative	Skill exch	Collabora	Positive		Positive	Positive	0	Low	No	High
RVW_021	Yusuf Abd	Freelance	Quora	2023-12-15		3	I tried Fre	Neutral	Peer colla	Collabora	Negative				1	Medium	No	Medium
RVW_022	Omar Has	Freelance	Quora	2025-07-02		5	I have bee	Positive	Skill exch	Collaboration		Positive	Negative	Positive	1	High	Yes	Low
RVW_023	Ahmed Su	Freelance	Quora	2025-05-18		5	I have bee	Positive	Learning t	Project-Ba	Positive	Negative		Negative	1	High	Yes	Low
RVW_024	Elena Ros	Freelance	Quora	2024-12-29		3	I tried Fre	Neutral	Project m	Project-Ba	Negative	Negative	Positive	Positive	1	Medium	No	Medium
RVW_025	Fatima Nc	Freelance	Quora	2024-06-04		5	I have bee	Positive	Skill exch	Skill Swap	Negative	Positive	Positive	Negative	1	High	Yes	Low
RVW_026	Aisha Bell	Freelance	Quora	2023-01-07		3	I tried Fre	Neutral	Skill exch	Project-Based Learn		Positive	Negative	Negative	1	Medium	No	Medium
RVW_027	Aisha Bell	Freelance	Quora	2024-04-17		5	I have bee	Positive	Team coll	Project-Based Learn		Positive			1	High	Yes	Low
RVW_028	Daniel Kir	Freelance	Quora	2023-09-14		1	My experi	Negative	Skill swap	Project-Based Learn		Positive	Positive	Negative	0	Low	No	High
RVW_029	Ahmed Su	Freelance	Quora	2025-12-04		1	My experi	Negative	Team coll	Collaboration		Negative	Positive	Negative	0	Low	No	High
RVW_030	Omar Has	Freelance	Quora	2023-01-23		5	I have bee	Positive	Peer colla	Collabora	Positive	Negative		Positive	1	High	Yes	Low

Image II: Dataset Before Data Transformation

Review ID	Reviewer Nam	App Na	Platform	Review D	Ra	Full Review Text	Sentimen	Mentioned Featu	Feature Ca	Featur	Featu	Featu	Payment	Trust Issu	UX Issu	Perform	Goal	User Sa	Would	Churn
RVW_001	Grace Thompson	Fiverr	PlayStore	6/23/2024	3	User feedback relat	Negative	Customer Support	UX						Negative		No	Low	No	High
RVW_002	Linda Perez	Fiverr	PlayStore	6/26/2024	5	User feedback relat	Negative	Customer Support	UX						Negative		No	Low	No	High
RVW_003	Aisha Lawal	Fiverr	PlayStore	12/2/2024	1	User feedback relat	Negative	Account Suspension	Trust					Negative			No	Low	No	High
RVW_004	Samuel Bello	Fiverr	PlayStore	7/25/2024	5	User feedback relat	Neutral	Gig Upload	Performance								No	Medium	Maybe	Mediur
RVW_005	Peter Johnson	Fiverr	PlayStore	4/30/2024	4	User feedback relat	Positive	Onboarding Experier	UX								Yes	High	Yes	Low
RVW_006	Fatima Musa	Fiverr	PlayStore	6/5/2024	2	User feedback relat	Negative	Account Suspension	Trust					Negative			No	Low	No	High
RVW_007	John Ade	Fiverr	PlayStore	5/6/2025	5	User feedback relat	Positive	Push Notifications	Notifications	UX							Yes	High	Yes	Low
RVW_008	Daniel Okorie	Fiverr	PlayStore	4/26/2024	5	User feedback relat	Positive	App Stability	Performance								Yes	High	Yes	Low
RVW_009	Mary Okafor	Fiverr	PlayStore	8/29/2024	3	User feedback relat	Neutral	Push Notifications	Notifications	UX							No	Medium	Maybe	Mediur
RVW_010	Kelvin Obi	Fiverr	PlayStore	7/16/2024	4	User feedback relat	Positive	Gig Upload	Performance								Yes	High	Yes	Low
RVW_011	Ruth Williams	Fiverr	PlayStore	8/13/2024	5	User feedback relat	Negative	Account Suspension	Trust					Negative			No	Low	No	High
RVW_012	Michael Chen	Fiverr	PlayStore	#########	4	User feedback relat	Positive	App Usability	UX								Yes	High	Yes	Low
RVW_013	u/techsavvy01	Fiverr	Reddit	1/12/2024	4	Used Fiverr to get a	Positive	Gig delivery	Service				Negative	Negative	Negative	Negative	Yes	High	Yes	Low
RVW_014	u/startupmind	Fiverr	Reddit	2/3/2024	3	Fiverr is good for qu	Neutral	Seller quality	Marketplace				Negative	Negative	Negative	Negative	Partial	Medium	Maybe	Mediur
RVW_015	u/anon_designer	Fiverr	Reddit	2/18/2024	2	Had issues with a fr	Negative	Customer Support	Support				Positive	Positive	Negative	Negative	No	Low	No	High
RVW_016	u/devtalks	Fiverr	Reddit	3/1/2024	5	Great experience hi	Positive	Milestones	Project Management				Negative	Negative	Negative	Negative	Yes	High	Yes	Low
RVW_017	u/uxcritic	Fiverr	Reddit	3/10/2024	3	The app works fine	Neutral	Interface	UX				Negative	Negative	Positive	Negative	Yes	Medium	Maybe	Mediur
RVW_018	u/freelancebuyer	Fiverr	Reddit	3/22/2024	4	Payment protectior	Positive	Payment protection	Trust	Safety			Negative	Negative	Negative	Negative	Yes	High	Yes	Low
RVW_019	u/complaintbox	Fiverr	Reddit	4/5/2024	1	Account got flaggec	Negative	Account moderatior	Trust	Safety			Negative	Positive	Negative	Negative	No	Low	No	High
RVW_020	u/contentcreator	Fiverr	Reddit	4/14/2024	5	Perfect for outsour	Positive	Gig marketplace	Marketplace				Negative	Negative	Negative	Negative	Yes	High	Yes	Low
RVW_021	u/bugreporter	Fiverr	Reddit	4/27/2024	2	Experienced slow lc	Negative	App Stability	Performance				Negative	Negative	Negative	Positive	No	Low	No	High
RVW_022	u/reviewreader	Fiverr	Reddit	5/2/2024	4	Reviews and ratings	Positive	Ratings system	Trust	Safety			Negative	Negative	Negative	Negative	Yes	High	Yes	Low
RVW_023	u/sidehustler	Fiverr	Reddit	5/11/2024	3	Fees can add up, bu	Neutral	Service fees	Pricing				Negative	Negative	Negative	Negative	Partial	Medium	Maybe	Mediur
RVW_024	u/brandbuilder	Fiverr	Reddit	5/19/2024	5	Found a long-term (	Positive	Talent discovery	Marketplace				Negative	Negative	Negative	Negative	Yes	High	Yes	Low
RVW_025	u/securepay	Fiverr	Reddit	6/1/2024	2	Refund process was	Negative	Refunds	Payments				Positive	Negative	Negative	Negative	No	Low	No	High
RVW_026	u/remotehire	Fiverr	Reddit	6/10/2024	4	Easy to find freelan	Positive	Search filters	UX				Negative	Negative	Negative	Negative	Yes	High	Yes	Low
RVW_027	u/researchthrow	Fiverr	Reddit	6/21/2024	3	Decent platform ov	Neutral	Seller vetting	Trust	Safety			Negative	Negative	Negative	Negative	Partial	Medium	Maybe	Mediur
RVW_028	David Miller	Freelancer	Quora	5/28/2023	1	My experience with	Negative	Skill exchange	Collaboration				Positive	Positive			No	Low	No	High
RVW_029	Daniel Kim	Freelancer	Quora	4/3/2023	3	I tried Freelancer.cc	Neutral	Skill swapping	Collaboration				Positive	Positive	Positive	Negative	Yes	Medium	No	Mediur

Image III: Dataset After Data Transformation

PRE-ANALYSIS

This section outlines the variables used in the project, their analytical roles, and how they support insight generation and predictive modeling.

1. Independent Variables

Independent variables represent the input features used to analyze and predict user behavior patterns across skill-swap and marketplace platforms.

- Rating – Numerical score provided by the user.
- Full Review Text – Unstructured textual feedback from users.
- Mentioned Feature – Specific feature referenced in the review.
- Feature Category 1–4 – Classified feature themes (e.g., UX, Payment and Trust).
- Payment Issue Flag – Binary indicator of payment-related complaints.
- Trust Issue Flag – Binary indicator of trust or safety concerns.

These variables serve as the primary drivers for sentiment analysis, satisfaction evaluation and churn prediction.

2. Project Split Variables

Split variables are used to segment, filter and contextualize the dataset for deeper analysis.

- UX Issue Flag – Indicates usability or design-related complaints.
- Performance Issue Flag – Captures speed, reliability or matching performance issues.
- App Name – Name of the reviewed application.
- Platform – Platform category or marketplace type.
- Review Date – Time dimension for trend and longitudinal analysis.

These variables enable comparative, temporal, and platform-level insights.

3. Dependent Variables

Dependent variables represent the outcomes the project aims to understand or predict.

- Sentiment Label – Emotional polarity derived from review text.
- Goal Achieved – Whether the user successfully met their intended objective.
- User Satisfaction – Overall satisfaction assessment.
- Would Recommend – Likelihood of recommending the platform.

- Churn Risk – Probability of user disengagement.

These outputs directly support retention strategy and performance optimization decisions.

DASHBOARD ANALYSIS & INTERPRETATION

This dashboard provides a consolidated analysis of 191 user reviews collected across multiple digital platforms. The objective is to uncover key experience pain points, assess user satisfaction trends and identify major drivers of churn risk to support product and platform improvement decisions.

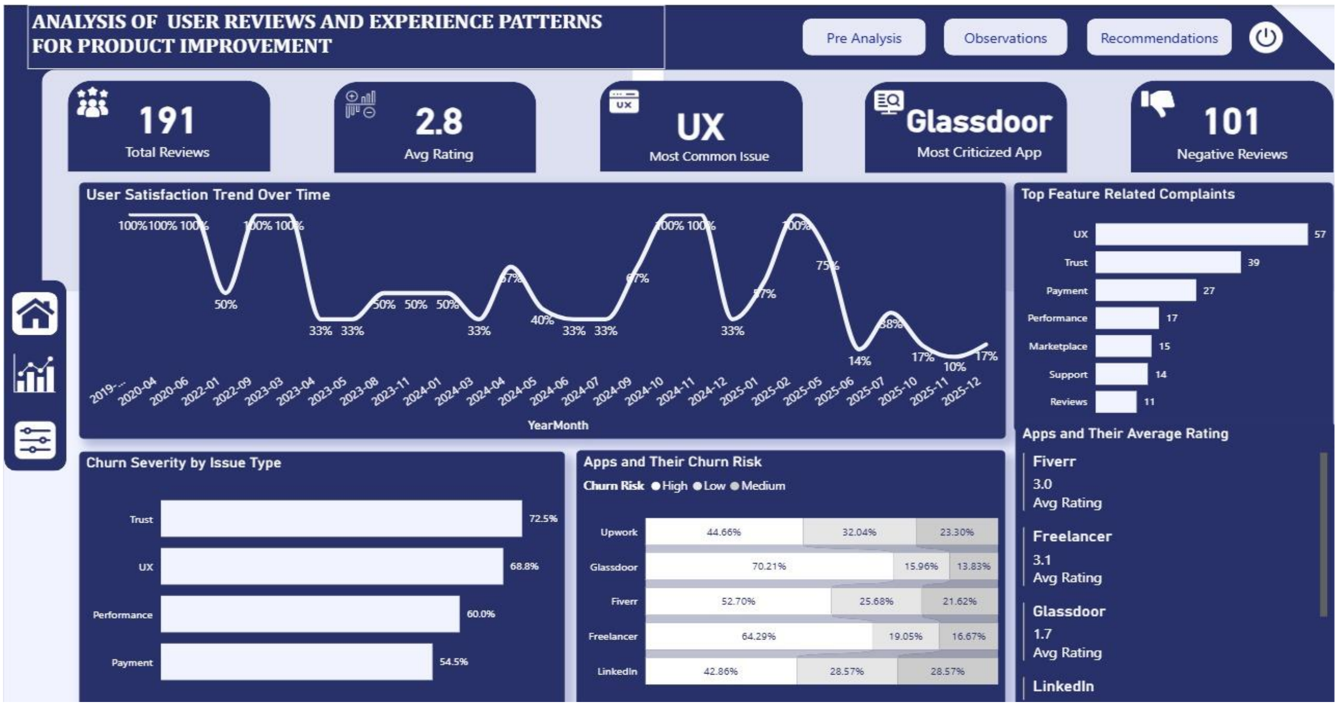

Image IV: Users Reviews and Experience Dashboard Screenshot

Image V: Users Reviews And Experience Dashboard Screenshot

Key KPIs Summary

- Total Reviews: 191
- Average Rating: 2.8
- Most Common Issue: UX
- Most Criticized App: Glassdoor
- Negative Reviews: 101

These headline metrics immediately indicate low overall satisfaction and a high volume of negative feedback, signaling a fragile market experience.

OBSERVATIONS & INSIGHTS

This section presents key observations and insights derived from the analysis of user reviews and experience patterns across competing skill-swap platforms.

1. Overall User Satisfaction and Market Health

Analysis of 191 user reviews across competing platforms shows an average rating of 2.8, indicating generally low user satisfaction within the skill-swap market. Sentiment distribution further reinforces this finding, with negative reviews (101) significantly outweighing positive reviews (52). In addition, the overall churn risk rate of 52.36% suggests that more than half of users are likely to disengage from existing platforms, highlighting weak market health and unmet user expectations.

2. User Experience (UX) as the Primary Source of Dissatisfaction

User experience-related challenges emerge as the most frequently reported complaint category across all analyzed platforms. UX issues contribute to approximately 68.8% of churn severity, making them a major driver of user dissatisfaction. Trends observed in the *User Satisfaction Over Time* analysis reveal recurring declines in satisfaction, particularly during onboarding stages and following platform updates, indicating poor usability design and inadequate change management.

3. Trust and Safety as the Strongest Churn Drivers

Trust and safety concerns exhibit the highest churn severity at 72.5%, surpassing all other issue categories. Regardless of platform-specific average ratings, trust-related complaints consistently align with elevated churn risk. Further sentiment analysis by platform shows that applications with a higher volume of trust issues experience disproportionately negative user sentiment, emphasizing trust as a critical determinant of platform sustainability.

4. Payment System as a Source of Friction

Payment-related challenges account for 54.5% churn severity and appear frequently in user complaints. Common issues include delayed payments, unclear fee structures, and frozen or restricted funds. These problems often overlap with trust-related concerns, compounding their negative impact on user sentiment and significantly reducing user retention.

5. Performance and Reliability Issues

Performance and reliability problems contribute to approximately 60% churn severity. Users report slow response times, unstable platform behavior, and inefficient or inaccurate skill and job matching mechanisms. When performance issues occur alongside UX or payment challenges, overall dissatisfaction intensifies, accelerating user disengagement.

6. Matching Transparency and Fairness

Market place-related complaints highlight strong user dissatisfaction with visibility, opportunity distribution, and opaque matching algorithms. Many users perceive existing systems as unfair, particularly where pay-to-win bidding models dominate visibility and access to opportunities. These models are consistently associated with negative sentiment and reduced trust, as users feel merit and skill quality are secondary to financial advantage.

7. Platform-Level Comparative Findings

Comparative analysis across platforms reveals significant performance gaps:

- Glassdoor records the lowest average rating (1.7) and the highest churn risk, indicating severe dissatisfaction.
- Freelancer is identified as the least recommended platform, driven by persistent UX and trust-related complaints.
- Upwork, despite strong brand recognition, continues to experience high churn risk linked to trust and payment concerns.
- Fiverr performs relatively better in overall satisfaction but still faces notable marketplace fairness and visibility challenges.

These findings suggest that brand strength alone is insufficient to sustain user retention without strong experience fundamentals.

8. Rating Discrepancy Versus User Sentiment

The analysis reveals that moderate numerical ratings (3–4 stars) do not necessarily equate to positive user sentiment. Hidden churn risk analysis shows that dissatisfaction often persists even at higher rating levels. This indicates that traditional rating systems may mask deeper frustrations, making sentiment analysis essential for accurately assessing platform health.

9. Support-Related Issues

Although support-related complaints occur less frequently than other issue categories, they are linked to exceptionally high churn rates when they do occur. Users experiencing unresolved support challenges often disengage immediately, suggesting that ineffective customer support acts as a critical tipping point in user abandonment.

RECOMMENDATIONS

1. **User Experience (UX) – Top Priority Intervention**

User experience emerged as the most critical driver of dissatisfaction and churn across reviewed platforms. To address this, onboarding processes should be significantly simplified by reducing unnecessary steps and adopting progressive disclosure, where complexity is introduced only when required. Role-based onboarding flows should be implemented to differentiate between new and experienced users, ensuring relevance and ease of use. In-product guidance such as tooltips, walkthroughs, and contextual help will further reduce user friction. At a strategic level, UX should be treated as a leadership-owned metric, supported by mandatory usability testing before major releases. UX-specific KPIs, including task completion time, error rates, and abandonment rates, should be tracked continuously. Micro-surveys should be deployed at critical user journeys, and feedback should be actively integrated into sprint planning rather than addressed only after release.

2. **Trust and Safety – Critical Risk Mitigation**

Trust-related issues pose a significant risk to user retention and platform credibility. Clear and accessible policies around dispute resolution, account enforcement, and fund handling should be published and consistently communicated. Users should receive real-time updates on disputes, verifications, and escalations to reduce uncertainty and frustration. Identity verification and fraud detection mechanisms must be strengthened, complemented by visible trust signals such as verified badges and transaction guarantees. A dedicated Trust & Safety escalation channel should be established, ensuring automated enforcement actions are auditable and supported by transparent appeal mechanisms with defined resolution timelines.

3. Payment Systems – Revenue and Retention Protection

Payment failures and ambiguities were strongly associated with negative sentiment and churn risk. To mitigate this, the platform should invest in redundant payment processing systems and monitor transaction latency and failure rates in real time. Users must be proactively informed of delays or issues, while fee structures should be simplified and clearly presented upfront. Detailed transaction breakdowns using plain language will help eliminate confusion and perceived hidden charges. Additionally, guaranteed resolution SLAs for payment disputes should be introduced, and support teams should be empowered to resolve payment-related issues without unnecessary escalation bottlenecks.

4. Performance and Reliability – Foundational Platform Stability

Platform performance and reliability form the foundation of user trust and satisfaction. Comprehensive performance and load audits should be conducted regularly, with priority given to uptime, latency, and scalability over rapid feature expansion. Automated monitoring and incident response systems should be implemented to ensure swift detection and resolution of issues. Improvements should also extend to job matching accuracy, with transparency provided on why certain matches are suggested. Continuous model retraining based on real user outcomes is essential, while uptime, response time, and failure rates should be tracked as executive-level KPIs.

5. Adopt a Proactive User Feedback and Retention Strategy

Given the high churn risk identified across similar platforms, a proactive approach to user feedback is essential. Continuous feedback loops, such as in-app surveys, exit surveys, and NPS tracking should be implemented to capture user sentiment in real time. Monitoring sentiment trends will help detect early warning signs of dissatisfaction, while churn prediction models can be used to identify at-risk users for targeted intervention. Communicating product improvements transparently demonstrates responsiveness, and rewarding long-term users through loyalty incentives or feature unlocks can further strengthen retention.

6. Align Product Roadmap with User Pain Points

To improve competitiveness and overall market health, product development efforts must be aligned with the most impactful user pain points. High churn-severity issues—particularly trust, UX, and performance, should take precedence over low-impact feature additions. KPIs should be refocused on user satisfaction and retention rather than growth alone, and regular benchmarking against competitors should be conducted to differentiate the platform on reliability, trust, and ease of use.

7. Improve Matching Transparency and Marketplace Fairness

Marketplace dissatisfaction is strongly linked to opaque matching systems and perceived pay-to-win dynamics. To address this, the platform should clearly explain how matching and ranking algorithms work using simple, accessible language. Overreliance on paid bidding should be reduced by balancing visibility with merit-based factors such as relevance, reliability, and performance history. Fair exposure mechanisms should be introduced to prevent opportunity monopolization, while visibility insights, such as why a user was or was not matched, will help manage expectations and reduce frustration.

CONCLUSION

This analysis of user reviews and experience patterns across comparable platforms provided valuable insights into the key factors influencing user satisfaction, recommendation behavior, and churn risk. The findings highlight that user experience, trust and safety, payment reliability, and platform performance are the most critical drivers of negative sentiment and disengagement.

By examining how users interact with existing apps similar to the proposed Skill Swap platform, this project moves beyond surface-level ratings to uncover underlying pain points that directly impact retention and long-term platform success. The structured use of sentiment labels, issue flags, and churn risk indicators enabled a data-driven assessment of where users struggle most and where product improvements will deliver the greatest impact.

The insights and recommendations from this analysis serve as a strategic foundation for building a more user-centric, trustworthy, and reliable platform. By proactively addressing the challenges identified in competing apps, the proposed product can avoid common pitfalls, improve user confidence, and create a differentiated experience that prioritizes usability, transparency, and long-term user retention.

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